

Efficient Quantum Algorithms for One-Dimensional Infrastructures

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Motivation and overview

- infrastructures are group-like structures
- they appear in arithmetic geometry in the study of computational problems related to number fields and function fields over finite fields
- arithmetic geometry provides a unified treatment of global fields
- the most prominent computational tasks are computing
 - the circumference of the infrastructure and
 - generalized discrete logarithms
- both problems are not known to have efficient classical algorithms

Discrete Infrastructures

- there is a simple and efficient quantum algorithm for solving these problems for infrastructures arising from function fields
- this is because such infrastructures are discrete and can be embedded into finite abelian groups
- we can simply use the efficient quantum algorithm for the abelian hidden subgroup problem to solve these two problems and also many other problems related function fields

1-dim nondiscrete infrastructures

- $\Rightarrow$ we focus on non-discrete infrastructures
- such infrastructures arise from number fields
- in this talk, I only consider one-dimensional infrastructures
- such infrastructures arise from number fields of unit rank

Improvements

- quadratic number fields give rise to one-dimensional infrastructures $\Rightarrow$ our algorithms can be used to solve Pell's equation and the principal ideal problem
- in this sense, they generalize Hallgren's quantum algorithms for these problems
- our improvements and extensions:
 - give a conceptually simpler presentation
 - find algorithms with lower complexity and constant success probability
 - introduce a technical result for analyzing algorithms based on Fourier sampling
 - present a more efficient method for computing in one-dimensional infrastructures

One-dimensional infrastructure

- a one-dimensional infrastructure of circumference R is a pair (X, d) where

- $X = \{x_0, x_1, \dots, x_{m-1}\}$ is a finite set and

- $d : X \rightarrow \mathbb{R}/R\mathbb{Z}$ is injective

- the elements of X are ordered such that

$$0 = d(x_0) < d(x_1) < \dots < d(x_{m-1}) < R$$

we use representatives in $[0, R)$ to order the $d(x_i)$

- imagine that we place the elements around a circle of circumference R starting at the top most point and moving clockwise

Baby step

- the baby step $\text{bs} : X \rightarrow X$ is defined as

$$\text{bs}(x_i) = \begin{cases} x_{i+1} & 0 \leq i < m - 1 \\ x_0 & i = m - 1 \end{cases}$$

it gives you the next element on the circle

- the relative distance function $\Delta_{\text{bs}} : X \rightarrow [0, R)$ is such that

$$d(\text{bs}(x)) = d(x) + \Delta_{\text{bs}}(x)$$

holds for all $x \in X$

it tells you how far you have to travel along the circle to get to the next element

Giant step

- for $x, y \in X$ consider the set

$$S_{x,y} = \{s \in [0, R) \mid d(x) + d(y) + s \in d(X)\}$$

- the giant step $\text{gs} : X \times X \rightarrow X$ is defined as

$$\text{gs}(x, y) = z$$

such that

$$d(z) = d(x) + d(y) + \min S_{x,y}$$

- the relative distance function $\Delta_{\text{gs}} : X \times X \rightarrow [0, R)$ is defined as

$$\Delta_{\text{gs}}(x, y) = \min S_{x,y}$$

Computational problems

- estimate the circumference R
- given $x \in X$, estimate $d(x)$
- these problems are computationally hard for classical computers
- the first generalizes the problem of computing the order of a (black-box) cyclic group
- the second generalizes the problem of determining discrete logarithms in a (black-box) cyclic group

Finite cyclic groups

- suppose $G = \langle g \rangle$ is a finite cyclic group of order R
- $\Rightarrow$ we can form an infrastructure out of G as follows:
 - $X = G$
 - $d(h) = \log_g(h)$
 - the baby step corresponds to multiplication by g
 - the giant step corresponds to the multiplication of two elements in G
 - Δ_{bs} always takes on the value 1
 - Δ_{gs} always takes on the value 0

Group arithmetic with f -reps

- a pair $(x, f) \in X \times \mathbb{R}$ is called an f -representation if

$$0 \leq f < \Delta_{\text{bs}}(x)$$

- denote the set of all f -representations by $\text{Rep}(\mathcal{I})$
- $\text{Rep}(\mathcal{I})$ can be endowed with a group structure such that

$$\text{Rep}(\mathcal{I}) \cong \mathbb{R}/R\mathbb{Z}$$

HSP over $\mathbb{R}$

- we have a surjective group homomorphism

$$h : \mathbb{R} \rightarrow \text{Rep}(\mathcal{I})$$

with

$$\ker h = R\mathbb{Z}$$

- $\Rightarrow$ computing the circumference corresponds to determining the subgroup of $\mathbb{R}$ hidden by h
- this is the intuition behind our algorithm
- but computing h is impossible due to finite precision
- this is because we can only approximate Δ_{bs} and Δ_{gs}

Computable version h_N of h

- Let $h_N : \mathbb{Z} \rightarrow X \times \mathbb{Z}$ be defined as

$$h_N(i) = (x, \lfloor fN \rfloor)$$

where s is some random offset and $h(i/N + s) = (x, f)$

- we can compute $h_N(i)$ exactly for $i \in W = \{0, \dots, q-1\}$
- for more than half $(x, k) \in f(W)$, the corresponding preimage has the special form

$$f^{-1}(x, k) = \{\lfloor k + jNR \rfloor \mid j \in \mathcal{J}\}$$

$\lfloor \cdot \rfloor$ is a function assigning to each $u \in \mathbb{R}$ either $\lfloor u \rfloor$ or $\lceil u \rceil$

- $\Rightarrow$ evaluating h_N in superposition over W and measuring induces a pseudo-periodic state

Pseudo-periodic states

- a quantum state in $\mathbb{C}^q$ is called ϵ -pseudo-periodic with period S if it is of the following form

$$|\psi\rangle = \frac{1}{\sqrt{|\mathcal{J}|}} \sum_{j \in \mathcal{J}} |[k + jS]\rangle$$

- $p = \lfloor q/S + 1 \rfloor$
 - $\mathcal{J} \subseteq \{0, 1, \dots, p-1\}$
 - $|\mathcal{J}| \geq \epsilon p$
 - $k \in \{0, 1, \dots, \lfloor S \rfloor\}$
- we can find a rational number $\hat{S}$ such that $|\hat{S} - S| \leq 1$ by Fourier sampling ϵ -pseudo-periodic states

Fourier sampling

- the lemma “perturbed geometric sum with missing terms” is at the heart of the analysis
 - let $\delta \in (-1, 1)$
 - ω be the p th root of unity $e^{2\pi i/p}$
 - θ an arbitrary real-valued function defined on $\mathcal{J} \subseteq \{0, 1, \dots, p-1\}$ with $|\theta_j| \leq \frac{p}{32}$
 -

$$|\mathcal{J}| \geq \frac{1 - \text{sinc}(\pi\delta)}{1 - 2 \sin(\pi/32)} p$$

$$\Rightarrow \frac{1}{|\mathcal{J}|^2} \left| \sum_{j \in \mathcal{J}} \omega^{\delta j + \theta_j} \right|^2 \geq \left(1 - 2 \sin(\pi/32) - (1 - \text{sinc}(\pi\delta)) \frac{p}{|\mathcal{J}|} \right)^2$$

Estimating the period/circumference

● INPUT:

- an upper bound M on $S > 1$ and q an integer such that $M^2 \leq q < 2M^2$
- a pair of ϵ -periodic states in $\mathbb{C}^q$ with period S
- for each ϵ -pseudo-periodic state, apply a Fourier transform over $\mathbb{Z}_q$ and measure to obtain c and d
- compute the convergents c_i/d_i of c/d where $d_i \leq \lfloor q/32 \rfloor$
- return $\mathcal{L} = \left\{ [c_i q / c] \mid d_i \leq \lfloor q/32 \rfloor \right\}$ as candidates for S

Success probability

- our algorithm has constant success probability since $\mathcal{L}$ contains an element $\hat{S}$ with $|\hat{S} - S| < 1$ with probability $\Omega(1)$
- Hallgren's algorithm succeeds with only $\Omega(1/\log^4(M))$ and works only if $\log M \geq 256$

Thank you!